

## ARTIFICIAL NEURAL NETWORKS OPEN ENDED LAB

**Course Title:** ANN Lab

**Course Code:** COMP-341L

**Semester & Section:** 6<sup>th</sup> (Red)

**Course Teacher:** Ms.Shakeela Shaheen

**Total Marks:** 10

**Time:** 30 minutes

**Student Name:** \_\_\_\_\_

**Reg. No.:** \_\_\_\_\_

| Program Learning Outcome | Course Learning Outcome | Learning Domain |
|--------------------------|-------------------------|-----------------|
| PLO-4                    | CLO-2                   | P-4             |

**CLO-2: Modify** neural network parameters to observe and analyze performance

**PLO-3:** Design / Development of Solutions

### Q. No 01:

You are working as a Junior AI Engineer at a startup building a **Real-Time Waste Classification System**.

The company has built a CNN model that classifies images into:

- Plastic
- Paper
- Metal

However, the CEO reports:

**“The model performs very well on training data but poorly on new images.”**

You are given a partially completed notebook with:

- Dataset loaded (CIFAR-10 subset or custom waste dataset)

- A basic CNN model implemented
- Training pipeline ready

Your task is to:

Modify neural network parameters to improve generalization and analyze performance changes.

**GIVEN MODEL (Base Code Provided):**

```
model = tf.keras.Sequential([
    tf.keras.layers.Conv2D(32, (3,3), activation='relu',
        input_shape=(32,32,3)),
    tf.keras.layers.MaxPooling2D(2,2),
    tf.keras.layers.Conv2D(64, (3,3), activation='relu'),
    tf.keras.layers.MaxPooling2D(2,2),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dense(3, activation='softmax')
])

model.compile(optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy'])
```

## TASKS

### Task 1 (1 Mark) - Overfitting Diagnosis

Train the model for 10 epochs.

Then:

1. Plot training vs validation accuracy.
2. Identify whether the model is overfitting.
3. Write 2 lines explaining how you know.

### Task 2 (2 Marks) - Modify Network Parameters

You must improve validation performance by modifying **at least TWO of the following**:

- Add Dropout

- Change number of filters
- Add Batch Normalization
- Change optimizer
- Modify learning rate
- Increase/decrease Dense units

Retrain for 10 epochs.

Record:

| Model Version | Train Acc | Val Acc |
|---------------|-----------|---------|
| Original      |           |         |
| Modified      |           |         |

### **Task 3 (3 Marks) – Transfer Learning Enhancement**

Replace the CNN with a pretrained model:

Use:

- MobileNetV2 (include\_top=False)

Freeze base layers.

Add classification head.

Train for 5 epochs.

Record validation accuracy.

### **Task 4 (4 Marks) – Fine-Tuning Decision**

Unfreeze last 10 layers.

Reduce learning rate.

Train 3 more epochs.

Answer:

1. Did validation improve?
2. Why reduce learning rate during fine-tuning?

**Marking Rubric**

| Component                        | Marks |
|----------------------------------|-------|
| Overfitting diagnosis            | 1     |
| Parameter modification           | 2     |
| Transfer learning implementation | 3     |
| Fine-tuning reasoning            | 4     |